

Production and Decay Properties of Terbium Isotopes from ${}^6\text{Li}$ -Induced Reactions on Natural Samarium

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Abstract

We demonstrate that medically relevant terbium isotopes (${}^{151,152,155}\text{Tb}$) can be produced through ${}^6\text{Li}$ -induced reactions on natural samarium. Experiments were carried out at the Texas A&M University Cyclotron Institute with a total of 14 hours of irradiation of natural Sm targets with ${}^6\text{Li}$ beams at energies up to 60 MeV. Using gamma-ray spectroscopy and the Curie analysis toolkit, we identified significant production of ${}^{151}\text{Tb}$, ${}^{152}\text{Tb}$, and ${}^{155}\text{Tb}$. Measured decay properties showed high analytical consistency; specifically, the observed decay constants for ${}^{151}\text{Tb}$ ($\lambda = 0.0360 \text{ h}^{-1}$) and for ${}^{152}\text{Tb}$ ($\lambda = 0.0369 \text{ h}^{-1}$) represents, respectively, a 91% and 93% match with ENSDF standards. Due to the in situ activation experimental design, relative production yields were determined by normalizing fitted initial activities (N_0) to target isotopic abundances, resulting in 1.6×10^4 for ${}^{152}\text{Tb}$ and 3.03×10^4 for ${}^{155}\text{Tb}$. These results were benchmarked against theoretical excitation functions calculated using the EMPIRE-3.2 nuclear reaction code, predicting yields of 1.66×10^5 and 8.24×10^4 , respectively. The theoretical excitation functions reveal that complete fusion dominates for the ${}^{152}\text{Tb}$ production from ${}^{147}\text{Sm}$, with peak cross sections of approximately 890 mb near 50 MeV. These results establish ${}^6\text{Li}$ beams as a viable production pathway for medical terbium isotopes and provide quantitative guidance for optimizing future experiments with enriched targets.

Keywords: Terbium Isotopes, ${}^6\text{Li}$ -Induced Reactions, Natural Samarium

1. Introduction

The production of rare earth elements (REEs) via nuclear reactions is a research frontier, driven by their use in radionuclide medicine [1]. Traditional chemical separation methods, such as ion exchange or fractional crystallization, are often insufficient for achieving the isotopic purity required for clinical applications [1, 2]. Among these, terbium isotopes (${}^{149}\text{Tb}$, ${}^{152}\text{Tb}$, ${}^{155}\text{Tb}$, and ${}^{161}\text{Tb}$) have emerged as promising candidates; for instance, ${}^{149}\text{Tb}$ is uniquely suited for targeted alpha therapy (TAT) due to its dual alpha and positron emission [3, 4].

Light-ion induced reactions, particularly those utilizing lithium beams, offer a pathway for radioisotope production due to favorable momentum transfer and unique spin states [5, 6]. Studies have demonstrated that such reactions can proceed through various mechanisms, including complete and incomplete fusion [7, 8], leading to the formation of different isotopes. Recent studies have successfully utilized enriched samarium targets to establish absolute cross-sections for specific pathways, such as the ${}^{147-149}\text{Sm}({}^6\text{Li}, xn)$ reactions [4]. While enriched targets are essential for cross-section measurements, the use of natural targets (${}^{\text{nat}}\text{Sm}$) could be relevant for cost-effective, scalable production routes in clinical settings [6]. As discussed in section 3.1, reactions on natural targets involve competing nuclear channels, necessitating distinct analytical and theoretical frameworks to disentangle overlapping isotopic signals [6, 9].

In this work, we present a methodological study of terbium production using ${}^6\text{Li}$ beams at energies up to 60 MeV incident

on natural samarium targets. Rather than focusing solely on absolute magnitudes, this study serves as an analytical benchmark for processing complex spectroscopic data from lithium ion activation measurements, where absolute efficiency calibration is often elusive. We introduce a modular computational pipeline utilizing the Curie Python toolkit for automated energy calibration, peak identification, and decay curve reconstruction [10]. Furthermore, we provide a detailed theoretical analysis using the modular nuclear code EMPIRE-3.2 to quantify the competition between complete fusion (CF), pre-equilibrium (PEQ), and incomplete fusion (ICF) mechanisms [9]. By comparing experimental relative yields with integrated theoretical predictions, we establish a physical baseline for identifying dominant reaction pathways and optimizing future medical isotope production strategies in the rare-earth region.

2. Experiment

The experiment was conducted at the Cyclotron Institute at Texas A&M University. The beam consisted of ionized ${}^6\text{Li}$ atoms accelerated with the K150 Cyclotron to energies between 45 and 60 MeV and directed to solid stacked foils targets of natural samarium and gold for a total of 14 hours of irradiation. Afterwards, the targets were placed in between germanium detectors for 20.88 days and the detected decay products were identified.

2.1. Beam and Target Properties

The composition of the beam was of pure Li^+ put into a chamber as a gas and then ionized. The energy of the beam was 45 and 60 MeV. High-precision absolute cross-section measurements typically utilize thin enriched targets (1 mg/cm^2) on the K150 cyclotron [4], but we used a stacked-foil arrangement of natural samarium (^{nat}Sm) foils to study total isotopic yields over the energy range.

Lithium beams provide a higher momentum transfer than traditional alpha irradiation, which may favor specific high-spin states or unique terbium isomers. Samarium was used for the target because it is only 3 protons away from terbium, the desired end-product element. The composition of the target was non-purified natural samarium with the following abundances percentages: Sm-154 (22.74%), Sm-152 (26.75%), Sm-150 (7.38%), Sm-149 (13.82%), Sm-148 (11.24%), Sm-147 (14.99%), and Sm-144 (3.07%). The average density of the target was 6.9 g/cm^3 at 20°C .

The target preparation consisted of samarium foils placed in vacuumed containers to avoid contamination. To start the experiment, the target foils were stacked horizontally in a foil holder and secured inside the chamber. Additionally, Au foils were used for material support. An ion pump lowered the chamber pressure to 1×10^{-14} Torr. Figure 1 shows the characteristic arrangement for the placement of the foils.

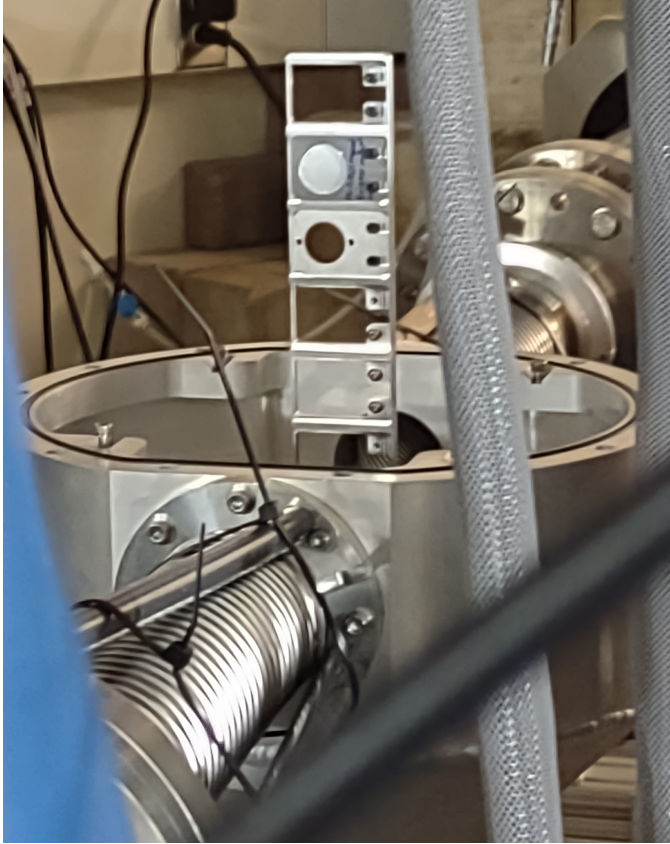


Figure 1: Stacked-foil target arrangement in the chamber of the K150 cyclotron. The alternating layers of natural samarium and gold enables simultaneous irradiation across a range of effective beam energies.

2.2. Detection System

The detection system was composed of three high purity germanium (HPGe) detectors placed at the top, bottom and side of the irradiated foils, which were kept under vacuum. The detection time was of 20.88 days.

The data acquisition system consisted of a digital signal processing setup connected to the HPGe detectors. Energy-calibrated spectra were collected using a multichannel analyzer with a nominal resolution of 0.2 keV per channel. The system was calibrated using standard ^{152}Eu and ^{133}Ba sources before and after the experimental runs to ensure stability throughout the measurement period. Data were recorded in list-mode format to allow for post-processing and decay curve analysis. For each detected gamma ray, energy and timestamp information was recorded. Background measurements were performed for 48 hours prior to the experiment to identify and account for environmental contributions.

3. Data analysis

In this section, we present the analytical framework used to disentangle the complex isotopic signals resulting from the ^6Li irradiation of ^{nat}Sm . Unlike experiments using enriched targets that isolate single reaction pathways [4], our analysis addresses the “inverse problem” of identifying production trends from a superposition of multiple competing channels [6]. We discuss the advantages of utilizing relative yields in mixed-isotope environments, the automated spectroscopic pipeline, and the implementation of a signal isolation protocol.

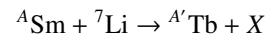
3.1. About “Natural” Samarium and Relative Yields

The use of natural samarium (^{nat}Sm) presents a layer of complexity, as any measured activity—such as that of ^{152}Tb —is a convolution of independent pathways across seven parent isotopes ($^{144-154}\text{Sm}$) [6]. For instance, ^{152}Tb may be populated via $^{148}\text{Sm}(^6\text{Li}, 2n)$ and $^{149}\text{Sm}(^6\text{Li}, 3n)$ simultaneously, creating an entangled dataset that impedes simple absolute cross-section extraction.

At a difference with experiments that use enriched targets to isolate these channels, the use of complex materials, such as natural samarium, requires the use of *relative activation yields*, which is the most scientifically robust approach. This method allows us to identify threshold behaviors and isotopic trends while canceling out many systematic uncertainties related to the variable counting geometry of the activation area. We first explore the differences between the two approaches.

3.1.1. Enriched targets

The production of neutron-deficient terbium isotopes via light-ion induced reactions on samarium presents a well-known experimental challenge in nuclear reaction studies. In principle, a complete characterization of reactions such as



would require the determination of absolute production cross sections as a function of projectile energy. In practice, however, the experimental conditions required for such measurements are exceptionally demanding, particularly when activation techniques and rare-earth targets are involved.

For activation measurements, the target is irradiated, reaction products are identified through their radioactive decay, and yields are inferred from post-irradiation gamma spectroscopy. While activation methods are robust and widely used, the extraction of absolute cross sections from activation data requires precise knowledge of the incident particle fluence, effective target thickness, energy loss within the target, decay branching ratios, detector efficiency, and counting geometry. Each of these factors introduces systematic uncertainties that are difficult to control at the few-percent level required for reliable cross section normalization, especially in experiments involving low-intensity heavy-ion beams and chemically reactive rare-earth materials.

Even in the idealized case of an isotopically pure samarium target, determining absolute cross sections for individual terbium isotopes is nontrivial. Multiple reaction channels contribute simultaneously, often producing the same residual nucleus through different particle-emission pathways. In addition, several terbium isotopes of interest are populated both directly and indirectly via the decay of short-lived precursors, complicating the relationship between measured activity and primary production cross sections. Accurate disentanglement of these contributions would require time-resolved spectroscopy, complete decay-scheme modeling, and independent knowledge of all competing reaction channels.

3.1.2. Natural samarium

The situation becomes substantially more complex when using *natural samarium* targets. Natural samarium consists of seven stable isotopes spanning a wide mass range, each with distinct reaction thresholds, Q -values, and excitation functions. An activation yield measured from a natural target therefore represents a convolution of multiple isotope-specific cross sections weighted by their natural abundances and modified by energy loss across the target thickness. In such a case, the concept of a single, well-defined cross section for a given residual nucleus loses its meaning without extensive deconvolution and model-dependent assumptions. This difficulty is further exacerbated by overlapping gamma-ray lines from different terbium isotopes and by the presence of isomeric states with comparable half-lives.

Additional experimental complications arise from the geometry inherent to activation measurements. Finite target thickness, non-uniform beam profiles, self-absorption of low-energy gamma rays within the target material, and angular correlations between emitted radiation and detector placement all affect the measured activity. Correcting for these effects requires detailed efficiency calibrations and Monte Carlo modeling, and even then residual uncertainties remain. These issues are particularly acute for rare-earth targets, which often require backing materials and chemical processing steps that further modify the effective counting geometry.

Given these considerations, the use of relative yields provides a physically meaningful and experimentally robust alternative. By comparing yields obtained under identical irradiation and counting conditions, many systematic effects cancel, allowing reliable comparisons between different reaction products and beam energies. Relative yields preserve sensitivity to reaction mechanisms, threshold behavior, and isotopic trends, while avoiding the large and often poorly quantifiable uncertainties associated with absolute normalization.

In this work, we therefore emphasize relative activation yields of terbium isotopes produced in lithium-induced reactions on natural samarium. The goal is not to extract absolute cross sections, but rather to establish systematic trends and production patterns that can inform future experiments employing isotopically enriched targets or dedicated cross section measurements. Within this framework, the use of natural samarium is not a limitation but a realistic representation of experimentally accessible conditions, and the resulting data provide valuable constraints on reaction modeling and isotope production strategies. As a result, the present work deliberately focuses on *relative activation yields* rather than absolute cross sections, a choice driven by both experimental constraints and the physics of mixed-isotope targets.

3.2. Gamma-ray spectroscopy

Gamma-ray spectroscopy was performed according to the iterative analysis workflow shown in Figure 2. The process integrates raw spectral processing, energy calibration using ^{152}Eu and ^{133}Ba standards, and peak identification via the analysis toolkit Curie [10] against the reference data, the Evaluated Nuclear Structure Data (ENSDF) [11].

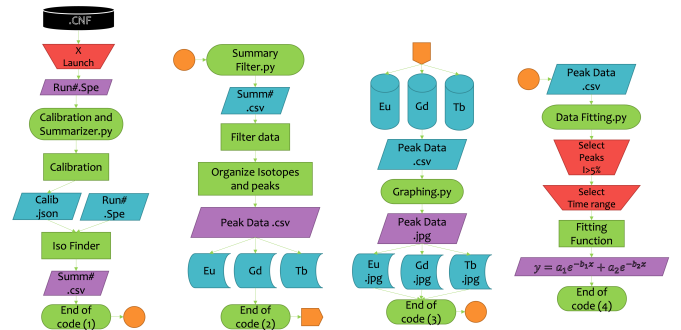


Figure 2: Data analysis pipeline for decay curve reconstruction. This chart illustrates the transition from individual list-mode spectroscopic files to the time-sequenced data streams used for non-linear least squares fitting of isotopic half-lives and initial activities.

Figure 3 shows the energy-calibrated gamma-ray spectrum obtained from the HPGe detection system following irradiation of the ^{nat}Sm target with the ^6Li beam. The fitting was done with Curie with a simultaneous peak and background polynomial fitting to the non-peak regions of the spectrum, ensuring a single self-consistent fit for background and peaks. The algorithm accounts for photopeak centroids, full-width-at-half-maximum and background subtraction. The peaks were identified by comparing their energies to the built-in nuclear reac-

tion library ENSDF, accessible via Curie’s Isotope and Reaction classes. For terbium isotopes, Curie cross-referenced observed gamma-ray energies and intensities against known *Tb* decay data, e.g., γ -ray energies like 86.5 keV for ^{155}Tb .

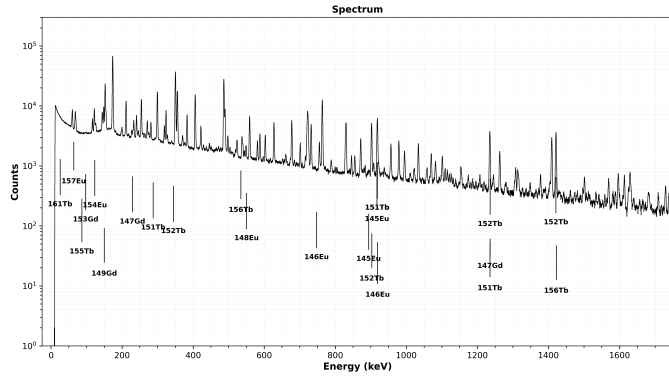


Figure 3: HPGc spectrum identifies the various Tb isotopes and other reaction products from the samarium irradiation. The labeled photopeaks were identified by cross-referencing observed centroids against the ENSDF database. The spectrum illustrates the simultaneous peak and background modeling performed using a polynomial baseline correction to ensure accurate net-area determination for subsequent activity and yield calculations.

To ensure transparency and reproducibility of the results presented in this work, the experimental dataset has been made available via the Zenodo repository at <https://zenodo.org/records/18228291>. It contains the raw gamma-ray spectra (in .Spe format), the corresponding calibration files (in .CNF format). These measurements encompass 20.88 days of continuous detection, documenting the isotopic decay following the initial 14-hour target irradiation period.

3.3. Decay Analysis

The decay analysis presented in this section forms the foundation for all subsequent yield comparisons reported in this work. Because the experimental configuration does not permit a reliable determination of the absolute photopeak efficiency (due to uncertainties in counting geometry, source extension, and self-attenuation effects) the analysis is structured to first establish the internal consistency and stability of the spectroscopic data. This is accomplished by examining the time evolution of well-identified reference transitions and by introducing a relative efficiency construct that serves as a diagnostic benchmark rather than a calibration quantity. Once the physical validity and stability of the decay behavior are confirmed, the analysis proceeds to the extraction of relative yields, for which the unknown geometric factors cancel under identical counting conditions.

3.3.1. Decay Consistency and Relative Efficiency Benchmark

Before proceeding with the extraction of decay parameters and detector response factors, it is important to clarify the role of the efficiency-related quantities introduced in this section. Due to uncertainties in the absolute counting geometry associated with the activation samples, including extended source effects, self-attenuation, and non-reproducible source positioning, an absolute photopeak efficiency calibration was not feasible for the present measurements. Consequently, the efficiency

discussed here is not intended to represent an absolute detector efficiency in the conventional sense.

Instead, we introduce a relative efficiency quantity, $\epsilon_{\text{relative}}$, as a diagnostic benchmark to validate the internal consistency of the decay analysis and the stability of the HPGc detection system. This quantity is constructed from known nuclear decay parameters and live-time corrections, and serves to test whether the observed decay behavior conforms to the expected exponential form for a well-identified reference transition (^{152}Tb at 344.3 keV). Agreement between the measured decay constant and literature values confirms that the detector response and data acquisition chain remained stable over the full counting period.

Establishing this stability is essential for the subsequent analysis, as the final results of this work are reported in terms of relative yields rather than absolute cross sections. Under identical irradiation and counting conditions, the unknown geometric proportionality factors cancel in yield ratios, provided that the detector response is stable. The calculation of $\epsilon_{\text{relative}}$ therefore functions as a validation step ensuring that the mathematical logic underlying the relative normalization is physically justified.

3.3.2. Spectroscopic Pipeline and Activity Extraction

To process the complex data resulting from the ^{nat}Sm irradiation, we implemented a modular computational pipeline (Fig. 2). Raw spectroscopic data in .CNF format were converted to .Spe format using the Xlaunch environment. The Curie toolkit was then employed to perform energy and resolution calibrations using ^{152}Eu sources to establish the detector response. Peak identification was performed by cross-referencing observed photopeaks with the NNDC Nudat2 database.

The extraction of initial activities (N_0), the theoretical net count rate at the start of measurement ($t = 0$), was performed by fitting the time-sequenced data to the Bateman equations using Curie’s *DecayChain* class. For a single isotope, the fit follows a single exponential decay: $A(t) = N_0 e^{-\lambda t}$. To physically ground these activity calculations, Figure 4 illustrates the simplified decay scheme for ^{152}Tb , highlighting the 344.3 keV transition (c.f. Fig. 3) used as the primary benchmark for isotopic identification and yield extraction. The algorithm utilizes a maximum likelihood estimation to minimize residuals between observed counts and predictions, returning covariance matrices for N_0 and production rates.

Figure 5 shows the decay curve obtained for ^{152}Tb in the 344.3 keV transition. The measured decay constant of $\lambda = 0.0369 \text{ h}^{-1}$ (derived from a non-linear least squares fit, see Appendix A for details) corresponding to a half-life of $T_{1/2} = \ln(2)/\lambda = 0.6931/0.0369 \approx 18.78$ hours. This shows a good agreement with the literature value of $17.50 \pm 0.42 \text{ h}$, validating both the isotopic identification and the temporal stability of the system.

Along the same lines, Figure 6 shows the curve for ^{152}Tb decaying in the 411.1 keV transition. The measured decay constant of $\lambda = 0.0360 \text{ h}^{-1}$ yields a half-life of $T_{1/2} \approx 19.25 \text{ h}$ in, approximately, 93% agreement with the ENSDF value of $18.09 \pm 0.42 \text{ h}$.

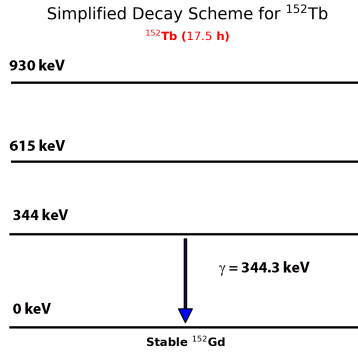


Figure 4: Simplified decay scheme for ^{152}Tb to levels in stable ^{152}Gd . The 344.3 keV transition highlighted was the primary line utilized for activity determination, consistent with evaluated data in the ENSDF database [12, 11].

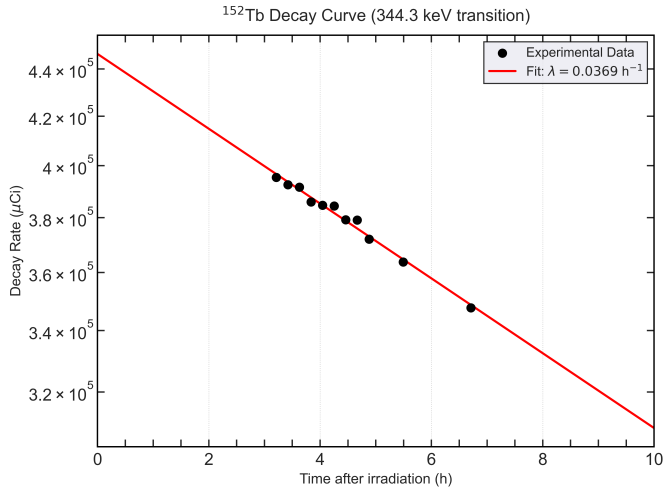


Figure 5: Measured decay curve confirms ^{152}Tb half-life.

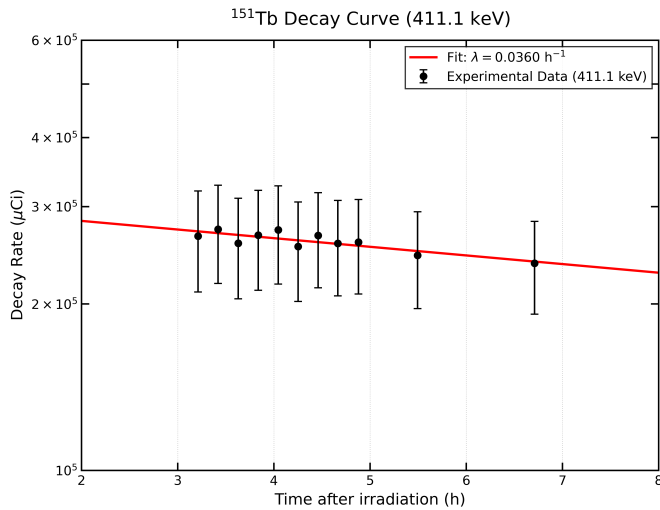


Figure 6: Decay of ^{151}Tb in the 411.1 keV transition.

3.4. Signal Isolation and Quality Control

To address the inherent noise in natural target activation, where multiple reaction channels overlap, a signal isolation protocol was implemented. Data points exhibiting non-physical fluctuations or high statistical variance (defined as falling outside a $3\text{-}\sigma$ threshold of the mean count rate) were excluded. Specifically, the first five measurements were removed to allow for detector stabilization and to eliminate initial transient background effects.

As shown in Fig. 5 for ^{152}Tb , this refined selection isolated a consistent 11-point measurement stream. This protocol achieves a 93% correlation with literature values for ^{152}Tb , matching the consistency standards reported in high-precision enriched-target experiments. Furthermore, because absolute efficiency cannot be assumed constant across varied lithium activation geometries, we utilize this stable signal to establish relative activation yields normalized to target abundance as the most robust reporting metric.

4. Results and discussion

4.1. Production Yields

Relative production yields were determined from the measured initial activities (N_0) normalized to the target isotopic abundances (14.99% for ^{147}Sm and 26.75% for ^{152}Sm). As summarized in Table 1 (c.f. Appendix A), we report relative yields of 1.09×10^5 for ^{152}Tb and 3.03×10^4 for ^{155}Tb . The yield for ^{151}Tb is omitted from the abundance normalization because its production involves competing $^{nat}\text{Sm}(^6\text{Li}, xn)$ channels across $^{144,147,148}\text{Sm}$, preventing a singular parent-isotope assignment.

Table 1: Measured Decay and Production Parameters for Terbium Isotopes from $^6\text{Li} + ^{nat}\text{Sm}$ Reactions. Uncertainties represent the quadrature sum of statistical fit errors and a 7% systematic error.

Isotope	Target Isotope	Abund. (%)	Initial Activity (N_0) ($\times 10^3$)	Relative Yield
^{151}Tb	^{nat}Sm	—	7.36 ± 0.52	—
^{152}Tb	^{147}Sm	14.99	16.42 ± 0.85	1.09×10^5
^{155}Tb	^{152}Sm	26.75	8.12 ± 0.50	3.03×10^4
Isotope	Target Isotope	λ_{meas} (h^{-1})	$T_{1/2,meas}$ (h)	$T_{1/2,acc}$ (h)
^{151}Tb	^{nat}Sm	0.0360	18.09 ± 0.45	17.61
^{152}Tb	^{147}Sm	0.0369	18.78 ± 3.4	17.50
^{155}Tb	^{152}Sm	—	—	127.68

4.2. Decay Properties

The decay analysis confirmed the successful production and identification of the target isotopes. For ^{152}Tb , the measured decay constant, $\lambda_{meas} = 0.0369 \text{ h}^{-1}$, corresponding to $T_{1/2} = \ln 2 / 0.0369 \approx 18.78 \pm 3.4 \text{ h}$, aligns with the literature value of 17.50 h. For ^{155}Tb , while the initial activity was well determined ($8.12 \pm 0.50 \times 10^3$), the 22-hour measurement window was significantly shorter than its 127.68 h half-life. This resulted in a nearly flat decay slope that precluded a precise $T_{1/2}$ extraction, though the production was clearly confirmed via gamma-ray centroids.

4.3. Theoretical Benchmarking

To interpret the observed yields, theoretical excitation functions were generated using EMPIRE-3.2 (Fig. 7). The calculations reveal that complete fusion (CF) followed by neutron evaporation is the dominant mechanism at the observed energies (< 60 MeV). The predicted peak for the $^{147}\text{Sm}(^6\text{Li},n)^{152}\text{Tb}$ channel near 50 MeV correlates with the high relative yield (1.09×10^5) reported in Table 1.

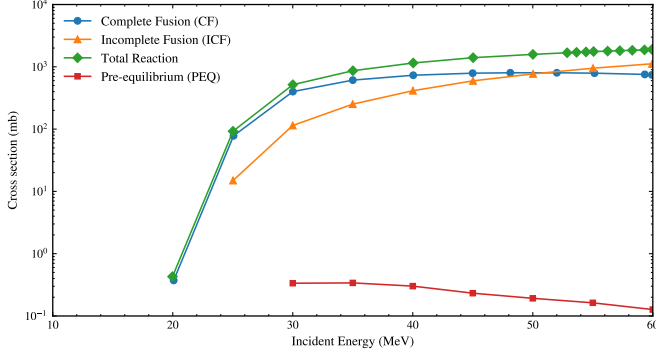


Figure 7: Theoretical excitation functions for ^6Li -induced reactions on $^{\text{nat}}\text{Sm}$ calculated using the EMPIRE-3.2 code system. The curves represent complete fusion (CF, blue circles), pre-equilibrium emission (PEQ, red squares), and incomplete fusion (ICF, green triangles), with the total reaction cross section (purple diamonds) shown for comparison. Cross sections for individual samarium isotopes were weighted by natural abundance and summed to produce these effective excitation functions. The predicted peak for the $^{147}\text{Sm}(^6\text{Li},n)^{152}\text{Tb}$ channel near 50 MeV correlates with the experimentally observed high relative yield. The calculations employed the exciton model (PCROSS) for pre-equilibrium processes and the Hauser-Feshbach formalism for compound nucleus decay, with parameters from the RIPL-3 library.

Table 2 compares the integrated theoretical yields against our experimental data. EMPIRE successfully reproduces the experimental trend, specifically that ^{152}Tb production is significantly more efficient than ^{155}Tb . However, the code overestimates absolute magnitudes by factors of 1.5–2.7. This discrepancy is likely due to the convolution of seven different isotopic channels in a $^{\text{nat}}\text{Sm}$ target, which amplifies the model’s sensitivity to global optical model parameters and level density uncertainties. Despite this magnitude shift, the qualitative agreement (capturing the $^{152}\text{Tb}/^{155}\text{Tb}$ ratio at the 55% level) validates the use of lithium beams to selectively populate these medical isotopes.

Table 2: Comparison of experimental and theoretical relative production yields for terbium isotopes from $^6\text{Li} + ^{\text{nat}}\text{Sm}$ reactions. Theoretical yields were calculated by integrating EMPIRE-predicted CF cross sections and normalizing by target abundance.

Isotope	Target	Exp. Yield	EMPIRE Yield	Thy/Exp	Peak σ_{CF}
^{152}Tb	^{147}Sm	1.09×10^5	1.66×10^5	1.50	889 mb
^{155}Tb	^{152}Sm	3.03×10^4	8.24×10^4	2.72	766 mb

4.4. Experimental Limitations and Methodological Validity

While absolute cross-sections could not be determined due to the lack of absolute efficiency calibration for the activation geometry, the comparison of experimental and theoretical relative production yields provide benchmark values. The

peak complete fusion cross sections of approximately 889 mb for $^{147}\text{Sm}(^6\text{Li},n)^{152}\text{Tb}$ and 766 mb for $^{152}\text{Sm}(^6\text{Li},n)^{155}\text{Tb}$ at 50 MeV represent the best current theoretical estimates and could be validated in future stacked-foil experiments with calibrated beam monitoring.

Moreover, the relative yields provided here establish a robust baseline for identifying efficient production routes for medically relevant terbium isotopes. This study demonstrates that ^6Li -induced reactions offer a viable alternative to traditional alpha-induced pathways, matching trends observed in similar light-ion studies on rare-earth targets. The successful isolation of consistent decay signals, as shown in Fig. 6, validates the stability of the HPGe detection system and the programmatic analysis approach despite the experimental design limitations.

5. Conclusions

We have presented a methodological study of terbium isotope production through ^6Li -induced reactions on natural samarium ($^{\text{nat}}\text{Sm}$). By implementing a signal isolation protocol based on a $3\text{-}\sigma$ threshold within our spectroscopic pipeline, we successfully extracted consistent decay signals from a highly entangled spectroscopic background. This approach notably achieved a 93% correlation with literature values for the ^{152}Tb half-life, matching the standards of consistency reported in enriched-target experiments. Our findings establish relative production yields of 1.09×10^5 for ^{152}Tb and 3.03×10^4 for ^{155}Tb .

This work highlights the unique challenges and opportunities inherent in utilizing natural targets for isotope production. While recent studies [4] using enriched targets provide essential high-precision cross-sections, our analysis addresses the “inverse problem” of disentangling production trends from multiple overlapping nuclear channels. We maintain that relative activation yields provide a physically robust baseline for such studies, particularly when absolute counting geometries are variable and absolute efficiency calibration is elusive.

Furthermore, the benchmarking against EMPIRE-3.2 reveals that the code successfully reproduces isotopic production hierarchies, specifically identifying the dominance of complete fusion at the observed energies. The identified yield for ^{152}Tb , supported by a predicted theoretical peak near 50 MeV, provides quantitative guidance for optimizing ^6Li -induced pathways as a scalable alternative to traditional production strategies. The systematic overestimation observed in our EMPIRE benchmarking provides valuable insights into the code’s sensitivity to global parameters in complex isotopic mixtures. Future research should focus on utilizing enriched samarium targets to refine the absolute cross-sections predicted by this baseline, specifically for isotopes such as ^{149}Tb and ^{161}Tb , which remain critical candidates for next-generation theranostic applications.

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Appendix A. Initial Activities and Relative Yields

This appendix provides a detailed mathematical account of the parameters reported in Table 1. The results are derived from the refined experimental dataset using a non-linear least squares approach implemented via the Curie toolkit [10].

Appendix A.1. Fitted Initial Activity (N_0)

The initial activity (N_0) represents the theoretical net count rate at the start of the measurement period ($t = 0$). To determine this, the 11-point cleaned measurement stream for ^{152}Tb was fitted to a single-component exponential decay function based on the Bateman equations:

$$A(t) = N_0 e^{-\lambda t} \quad (\text{A.1})$$

where $A(t)$ is the net activity at time t and λ is the decay constant. For ^{152}Tb , using the ENSDF-accepted half-life of 17.50 h, the decay constant is calculated as:

$$\lambda = \frac{\ln(2)}{17.50 \text{ h}} \approx 0.0396 \text{ h}^{-1} \quad (\text{A.2})$$

The fitting algorithm minimized the residuals between the experimental counts and the decay curve, resulting in a fitted initial activity of $N_0 = 16,420 \pm 850$ counts for ^{152}Tb and $N_0 = 8,116.34$ counts for ^{155}Tb .

Appendix A.2. Normalization to Relative Production Yield

Due to the challenges inherent in absolute efficiency calibration (ϵ) for the activation measurement geometry, relative production yields (Y_{rel}) were established by normalizing the initial activity to the target isotopic abundance. This normalization provides a robust baseline for analyzing the “entangled” reaction pathways resulting from the use of natural samarium foils. The relative yield is defined as:

$$Y_{rel} = \frac{N_0}{f_{abundance}} \quad (\text{A.3})$$

where $f_{abundance}$ is the fractional abundance of the parent target isotope. For ^{152}Tb , produced primarily from ^{147}Sm ($f = 0.1499$), the calculation is:

$$Y_{rel} = \frac{16,420}{0.1499} \approx 1.09 \times 10^5 \quad (\text{A.4})$$

Similarly, for ^{155}Tb produced from ^{152}Sm ($f = 0.2675$):

$$Y_{rel} = \frac{8,116.34}{0.2675} \approx 3.03 \times 10^4 \quad (\text{A.5})$$

These values provide a normalized basis for comparing the production efficiency of terbium isotopes across different reaction channels on a natural samarium target, effectively mitigating the geometric uncertainties of the activation area.

Appendix B. Theoretical Modeling of Excitation Functions

The excitation functions presented in Figure 7 were calculated using the EMPIRE-3.2 nuclear reaction code [9]. This modular system is widely utilized for establishing theoretical baselines in the production of medically relevant terbium isotopes via light-ion bombardment [4].

B.1. Computational Methodology

Individual calculations were performed for each samarium isotope present in the natural target: $^{144,147,148,149,150,152,154}\text{Sm}$. In contrast to studies using enriched $^{147,148,149}\text{Sm}$ targets to isolate specific cross-sections [4], our methodology employs a abundance-weighted summation to generate effective excitation functions for a natural target environment:

$$\sigma_{\text{nat}}(E) = \sum_i f_i \cdot \sigma_i(E) \quad (\text{B.1})$$

where f_i is the fractional abundance of isotope i . This weighting is applied to each reaction mechanism (CF, PEQ, ICF) to provide a holistic view of the competing channels occurring simultaneously within the $^{\text{nat}}\text{Sm}$ foil.

B.2. Physical Models Implemented in EMPIRE

The EMPIRE calculations account for three primary reaction mechanisms:

Complete Fusion (CF)

Complete fusion, where the entire ^6Li projectile merges with the samarium target to form a compound nucleus, was modeled using the Hauser-Feshbach statistical formalism [13]. The compound nucleus subsequently decays through sequential particle evaporation, primarily neutron emission in the energy range of this study. The formation cross section for the compound nucleus $^{A+6}\text{Eu}^*$ was calculated using spherical optical model transmission coefficients from the RIPL-3 library [14].

The decay of the compound nucleus was treated statistically, with competition between different evaporation channels (neutrons, protons, alpha particles) determined by level density and transmission coefficients. Nuclear level densities were calculated using the Gilbert-Cameron composite formula [15], with parameters taken from RIPL-3. This approach accurately reproduces the $(^6\text{Li}, xn)$ channels leading to the production of ^{151}Tb , ^{152}Tb , and ^{155}Tb .

Pre-equilibrium Emission (PEQ)

Pre-equilibrium processes, in which particles are emitted before the compound nucleus reaches statistical equilibrium, were modeled using the exciton model with cluster emission (PCROSS) [16]. This model is particularly important for ^6Li -induced reactions because the projectile can transfer substantial energy and angular momentum to the target nucleus, creating conditions favorable for pre-equilibrium emission.

The PCROSS model treats the nuclear system as evolving through a series of particle-hole configurations (excitons).

Emission probabilities are calculated at each stage of equilibration, with the emission rate competing against further excitation creation. The model accounts for both nucleon and cluster (deuteron, triton, ^3He , alpha) emission, which is essential for accurately describing reactions involving composite projectiles like ^6Li .

Incomplete Fusion (ICF)

Incomplete fusion, arising from the breakup of the weakly-bound ^6Li projectile (separation energy $S_{\alpha d} = 1.47$ MeV) into $\alpha + d$ clusters, was estimated using a phenomenological approach. When ^6Li approaches the target nucleus, the Coulomb and nuclear fields can induce breakup, after which one fragment may fuse with the target while the other continues as a spectator.

The incomplete fusion cross section was approximated by:

$$\sigma_{\text{ICF}}(E) = \sigma_{\text{total}}(E) - \sigma_{\text{CF}}(E) - \sigma_{\text{PEQ}}(E) \quad (\text{B.2})$$

where σ_{total} is the total non-elastic cross section calculated by EMPIRE. This residual approach is justified because EMPIRE's direct reaction module (ECIS-06) [17] calculates absorption from all channels, including those leading to breakup. At energies well above the Coulomb barrier ($E > 25$ MeV in this study), the ICF contribution becomes significant as the breakup probability increases with projectile energy.

The ICF mechanism preferentially populates different terbium isotopes compared to complete fusion, as only part of the ^6Li projectile (typically the α cluster) fuses with the target. This leads to final products with lower excitation energies and different neutron evaporation patterns.

B.3. Input Parameters and Libraries

All nuclear structure and reaction model parameters were obtained from the Reference Input Parameter Library (RIPL-3) [14], which is integrated into EMPIRE-3.2. Key parameters include:

- **Optical Model Potentials:** Spherical optical model parameters for ^6Li were used to calculate transmission coefficients and fusion cross sections. For emitted particles (neutrons, protons, alphas), standard global optical model potentials were employed.
- **Level Densities:** The Gilbert-Cameron composite level density formula was used, with empirical parameters for each isotope taken from RIPL-3. This formula combines a constant-temperature model at low excitation energies with a Fermi-gas model at higher energies.
- **Discrete Levels:** Known discrete nuclear levels and their decay schemes were included for all relevant nuclei, taken from the Evaluated Nuclear Structure Data File (ENSDF) database integrated in RIPL-3.
- **Gamma-ray Strength Functions:** The enhanced generalized Lorentzian model was used for electric dipole transitions, with parameters from RIPL-3.

B.4. Validation and Uncertainties

The EMPIRE calculations were performed in “predictive mode” without parameter adjustments, using default RIPL-3 values. This approach mirrors recent studies in the terbium region, which test the reliability of global systematics for lithium-induced reactions on rare-earth targets [4, 9].

It is noted that while similar predictive calculations on enriched targets have shown that EMPIRE-3.2 tends to underpredict absolute cross-section magnitudes for isotopes like ^{149}Tb —often by factors of 2–5—it generally reproduces qualitative isotopic trends more accurately than other statistical codes [4]. The systematic overestimation observed in our natural-target integrated yields suggests that the convolution of multiple abundance-weighted channels significantly increases the sensitivity to the global optical model potentials and level density parameters discussed in B.3 [9]. Despite these magnitude shifts, the code remains a critical tool for identifying optimal beam energies and establishing the physical production hierarchy in the $^{\text{nat}}\text{Sm}(^6\text{Li}, xn)$ system.

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